Day 8: Linux Terminal Fundamentals & Navigation

Phase 2: Linux & Computer Fundamentals | Focus: Terminal, Paths, Logs & Permissions

1. Linux Unified Directory Tree

Unlike Windows drive letters (C:\, D:\), Linux uses a single inverted tree starting at Root (/).
- /bin : Essential binary executables (ls, cd, python, cat).
- /etc : System-wide configuration files (nginx, sshd, network configs).
- /home : Personal user directories (/home/ubuntu, /home/alice).
- /var : Variable data files that change over time (system logs /var/log, databases).
- /tmp : Temporary runtime files (erased upon system reboot).

2. Absolute vs. Relative Paths

- Absolute Path: Begins at root (/) and provides the complete path regardless of your current location (e.g., /var/log/nginx/access.log).
- Relative Path: Evaluated relative to your Current Working Directory (CWD).
  - '.' : Represents current directory.
  - '..' : Represents parent directory (one level up).
  - '~' : Represents the current user's home directory (/home/username).

3. Core File Operations & Safe Log Inspection

- Navigation & Operations: pwd (print CWD), ls -la (detailed list including hidden files), cd, mkdir -p, touch, cp -r, mv, rm -r (recursive deletion).
- Log Viewing Strategy:
  - cat file.log : Dumps full file at once. DANGEROUS on large log files (floods terminal buffer).
  - head -n 10 file.log : Prints first 10 lines from top.
  - tail -n 10 file.log : Prints last 10 lines from bottom (ideal for checking recent events).
  - tail -f file.log : Follows file changes in real-time as new lines are appended.

4. Linux File Permissions & Ownership (chmod / chown)

ls -l displays permissions in 3 triplets: Owner (u), Group (g), Others (o).
- Values: Read (r = 4), Write (w = 2), Execute (x = 1).
- Numeric Permissions (chmod):
  - 644 (rw-r--r--) : Owner read/write; Group/Others read only. Standard for documents.
  - 755 (rwxr-xr-x) : Owner full control; Group/Others read & execute. Standard for scripts/dirs.
  - 600 (rw-------) : Owner read/write only; Others blocked. Mandatory for SSH private keys (id_rsa).
- Permission Denied Fix: Running chmod +x script.sh grants execute permission to the file.